

Practice Assignment #3: Proofs

1. Prove that in the code below, *count* computes the length of the linked list.

```
count = 0
curr = head

while curr != null:
    count++
    curr = curr.next
```

2. Prove that the code below reverses the array *A* of size *n*.

```
i = 0
j = n - 1

while i < j:
    swap(A[i], A[j])
    i = i + 1
    j = j - 1
```

3. Prove that a binary tree contains at most 2^d nodes at depth d . Then use this result to prove the maximum-node bound for a binary tree of height h .
4. Prove that in a binary tree with n nodes, exactly $n + 1$ child pointers are null.
5. Prove that in a binary search tree with all unique keys, it is never necessary to search both the left and right subtrees of a node.